

The Broken Food Web

II PUC · Ch 12 · Ecosystem

Starring Kiran 👦 & the Pond Guardians 🐸

A BAIO storybook by Narendra Sir · Biology Lecturer



Where Did the Frogs Go?

In a peaceful village pond, the frogs suddenly disappeared. Soon insects swarmed everywhere, water plants were eaten faster, fish found less food and the pond turned murky. Kiran and his friends, the Pond Guardians, decided to become ecosystem engineers. "First," said Kiran, "we must understand our **ecosystem** — living things and their surroundings working as one unit."

💡 **REMEMBER**

An ecosystem has biotic components (producers, consumers, decomposers) and abiotic components (light, water, soil, temperature).



The Food Makers

They began with the green life: tiny **phytoplankton**, algae and floating and rooted plants. "These are **producers** — they make food by photosynthesis," said Kiran. The rate at which they make biomass is **productivity**. Total food made is **gross primary productivity** (GPP); what remains after respiration is **net primary productivity** (NPP).

💡 **REMEMBER**

NPP = GPP − R (respiration losses). NPP is the biomass available to consumers.



Who Eats Whom?

Next came the eaters, or **consumers**. Insects and zooplankton ate plants — **primary consumers**. Frogs and small fish ate insects — **secondary consumers**. Snakes and herons ate frogs — **tertiary consumers**. Each feeding step is a **trophic level**. Arranged in a line, they make a **food chain** — the **grazing food chain** (GFC) that starts with producers.

💡 **REMEMBER**

The detritus food chain (DFC) begins with dead organic matter; in an aquatic ecosystem the GFC is the major conduit of energy flow.



The Web Unravels

Kiran drew arrows between all the animals. The chains crossed and joined into a **food web**. "Look — frogs ate insects and fed the snakes. Without frogs, insects multiplied and ate the plants, while snakes and herons went hungry." Removing one species had shaken the whole web. "Every organism depends on others," whispered the youngest Guardian.

💡 **REMEMBER**

In a food web, energy flows along many interconnected food chains.



The Clean-Up Crew

At the muddy edge, earthworms, bacteria and fungi were at work — the **decomposers**. Kiran listed the steps of **decomposition**: **fragmentation** by detritivores, **leaching** of soluble nutrients, **catabolism** by enzymes, **humification** forming dark **humus**, and **mineralisation** that releases nutrients back to the soil.

💡 **REMEMBER**

Decomposition is faster when it is warm and moist, and slower in cold and dry conditions or without oxygen.

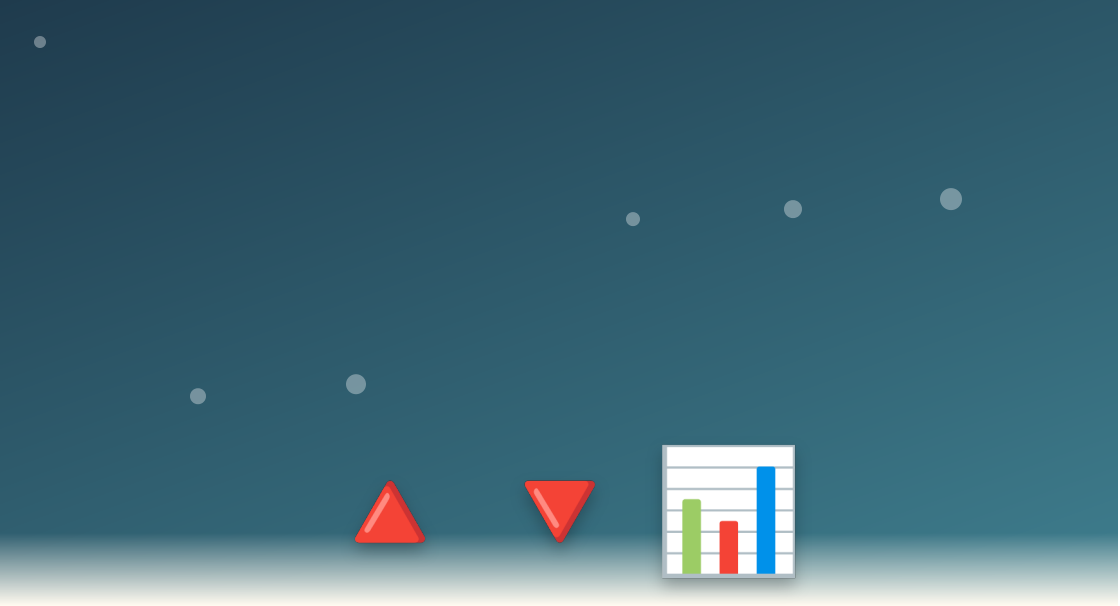


Following the Energy

"All energy comes from the sun," said Kiran, "but plants capture only 2–10 percent of the **photosynthetically active radiation** (PAR)." Energy flows one way — **unidirectional** — from producers to consumers. At each step only about 10 percent passes on; the rest is lost as heat. This is the **10 percent law** of **Lindeman**.

💡 **REMEMBER**

PAR is about 50 percent of the incident solar radiation.



Building Pyramids

The Guardians built **ecological pyramids** of number, biomass and energy. The pyramid of biomass in the sea is **inverted**, because the small fish outweigh the phytoplankton at any moment. "But the **pyramid of energy** is always upright," said Kiran, "because energy is always lost at every level." The mass of living matter at each level is the **standing crop**.

💡 **REMEMBER**

Pyramids do not include saprophytes, even though they play a vital role in the ecosystem.



The Frogs Return

The Guardians cleaned the pond and stopped pesticides from draining into it. Slowly the frogs came back. Insects decreased, plants recovered, fish grew fat and herons returned. Nutrients cycled again through the decomposers. "Balance is restored," smiled Kiran. "An ecosystem is like a web of threads — pull one, and the whole web trembles."

💡 **REMEMBER**

Ecosystem services such as pollination, purification of water and nutrient cycling are provided free by healthy ecosystems.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

ecosystem p.1 · phytoplankton p.2 · producers p.2 · productivity p.2 · gross primary productivity p.2 · net primary productivity p.2 · consumers p.3 · primary consumers p.3 · secondary consumers p.3 · tertiary consumers p.3 · trophic level p.3 · food chain p.3 · grazing food chain p.3 · food web p.4 · decomposers p.5 · decomposition p.5 · fragmentation p.5 · leaching p.5 · catabolism p.5 · humification p.5 · humus p.5 · mineralisation p.5 · photosynthetically active radiation p.6 · unidirectional p.6 · 10 percent law p.6 · Lindeman p.6 · ecological pyramids p.7 · inverted p.7 · pyramid of energy p.7 · standing crop p.7



QUICK QUIZ

Do you remember?

1. What is a producer?

- A plant-eating animal
- Makes its own food by photosynthesis
- A decomposer
- A top predator

2. What happens if frogs are removed from the pond food web?

- Nothing changes
- Insects increase; the web is unbalanced
- All plants grow better
- Energy flow increases

3. What is the role of decomposers?

- Make food from sunlight
- Break down dead matter, return nutrients
- Eat living plants
- Produce oxygen

4. According to Lindeman, energy passed to the next trophic level is about:

- 1%
- 10%
- 50%
- 90%

5. Which ecological pyramid is always upright?

- Number
- Biomass
- Energy
- None

Answers: 1-Makes its own food by photosynthesis · 2-Insects increase; the web is unbalanced · 3-Break down dead matter, return nutrients · 4-10% · 5-Energy